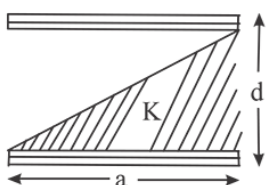


JEE Mains 2019 Chapter wise Question Bank

Capacitance - Questions

Q1

A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d ($d \ll a$). The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is:

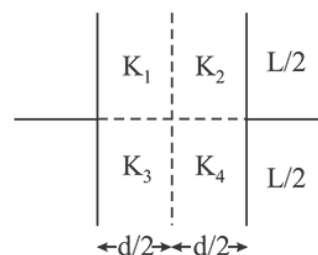


- (1) $\frac{K \epsilon_0 a^2}{2d(K+1)}$ (2) $\frac{K \epsilon_0 a^2}{d(K-1)} \ln K$
 (3) $\frac{K \epsilon_0 a^2}{d} \ln K$ (4) $\frac{1}{2} \frac{K \epsilon_0 a^2}{d}$

9 Jan Morning

Q2

A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1, K_2, K_3, K_4 arranged as shown in the figure. The effective dielectric constant K will be:



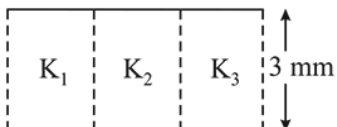
- (1) $K = \frac{(K_1 + K_3)(K_2 + K_4)}{K_1 + K_2 + K_3 + K_4}$
 (2) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{2(K_1 + K_2 + K_3 + K_4)}$
 (3) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{K_1 + K_2 + K_3 + K_4}$
 (4) $K = \frac{(K_1 + K_4)(K_2 + K_3)}{2(K_1 + K_2 + K_3 + K_4)}$

9 Jan Evening

Q3

Capacitance

A parallel plate capacitor is of area 6 cm^2 and a separation 3 mm . The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants $K_1 = 10$, $K_2 = 12$ and $K_3 = 1$ (4). The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be:



- (1) 4 (2) 14 (3) 12 (4) 36

10 Jan Morning

Q4

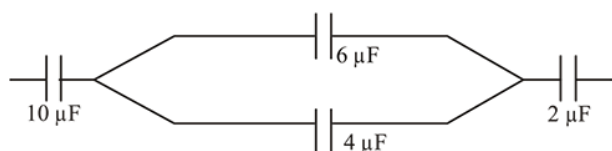
A parallel plate capacitor having capacitance 12 pF is charged by a battery to a potential difference of 10 V between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is:

- (1) 692 pJ (2) 508 pJ
(3) 560 pJ (4) 600 pJ

10 Jan Evening

Q5

In the figure shown below, the charge on the left plate of the $10 \mu\text{F}$ capacitor is $-30 \mu\text{C}$. The charge on the right plate of the $6 \mu\text{F}$ capacitor is :



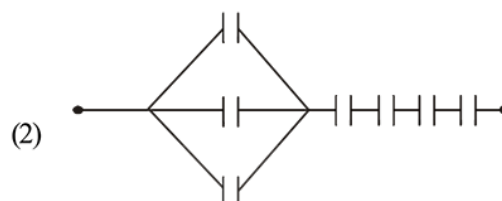
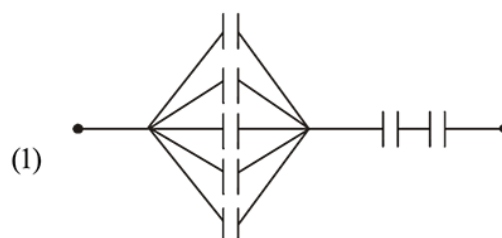
- (1) $-12 \mu\text{C}$ (3) $+12 \mu\text{C}$
(2) $-18 \mu\text{C}$ (4) $+18 \mu\text{C}$

11 Jan Morning

Q6

JEE Mains 2019 Chapter wise Question Bank

Seven capacitors, each of capacitance $2 \mu\text{F}$, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right) \mu\text{F}$. Which of the combinations, shown in figures below, will achieve the desired value?

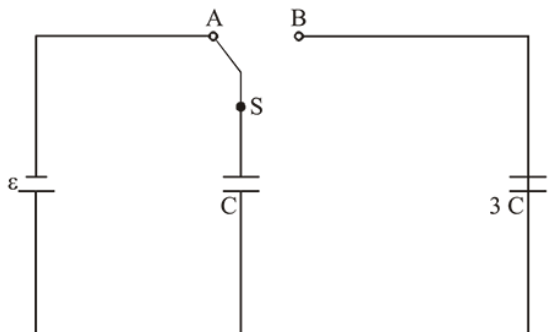


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Q7

Capacitance

In the figure shown, after the switch 'S' is turned from position 'A' to position 'B', the energy dissipated in the circuit in terms of capacitance 'C' and total charge 'Q' is:

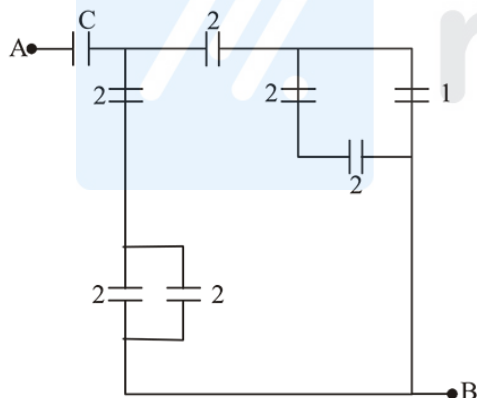


- (1) $\frac{1}{8} \frac{Q^2}{C}$ (2) $\frac{3}{8} \frac{Q^2}{C}$
 (3) $\frac{5}{8} \frac{Q^2}{C}$ (4) $\frac{3}{4} \frac{Q^2}{C}$

12 Jan Morning

Q8

In the circuit shown, find C if the effective capacitance of the whole circuit is to be $0.5 \mu\text{F}$. All values in the circuit are in μF .



- (1) $\frac{7}{11} \mu\text{F}$ (2) $\frac{6}{5} \mu\text{F}$
 (3) $4 \mu\text{F}$ (4) $\frac{7}{10} \mu\text{F}$

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Q9

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A parallel plate capacitor with plates of area 1 m^2 each, are at a separation of 0.1 m . If the electric field between the plates is 100 N/C , the magnitude of charge on each plate is :

(Take $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}-\text{M}^2}$)

- (1) $7.85 \times 10^{-10} \text{ C}$ (2) $6.85 \times 10^{-10} \text{ C}$
 (3) $8.85 \times 10^{-10} \text{ C}$ (4) $9.85 \times 10^{-10} \text{ C}$

12 Jan Evening

Q10

Voltage rating of a parallel plate capacitor is 500 V . Its dielectric can withstand a maximum electric field of 10^6 V/m . The plate area is 10^{-4} m^2 . What is the dielectric constant if the capacitance is 15 pF ?

(given $\epsilon_0 = 8.86 \times 10^{-12} \text{ C}^2 \text{ m}^2$)

- (1) 3.8 (2) 8.5 (3) 4.5 (4) 6.2

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Q11

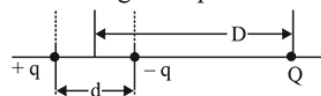
A parallel plate capacitor has $1 \mu\text{F}$ capacitance. One of its two plates is given $+2 \mu\text{C}$ charge and the other plate, $+4 \mu\text{C}$ charge. The potential difference developed across the capacitor is :

- (1) 3 V (2) 1 V (3) 5 V (4) 2 V

8 April Evening

Q11

A system of three charges are placed as shown in the figure:



If $D \gg d$, the potential energy of the system is best given by

- (1) $\frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{d} - \frac{qQd}{2D^2} \right]$ (2) $\frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{d} + \frac{2qQd}{D^2} \right]$
 (3) $\frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$ (4) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right]$

9 April Morning

Q12

A capacitor with capacitance $5 \mu\text{F}$ is charged to $5 \mu\text{C}$. If the plates are pulled apart to reduce the capacitance to $2 \mu\text{F}$, how much work is done?

- (1) $6.25 \times 10^{-6} \text{ J}$ (2) $3.75 \times 10^{-6} \text{ J}$
 (3) $2.16 \times 10^{-6} \text{ J}$ (4) $2.55 \times 10^{-6} \text{ J}$

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Q13

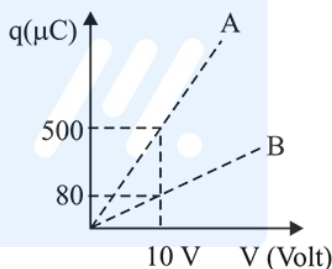
The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V . When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is:

- (1) $\frac{nV}{K+n}$ (2) V
 (3) $\frac{V}{K+n}$ (4) $\frac{(n+1)V}{(K+n)}$

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Q14

Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitances are :



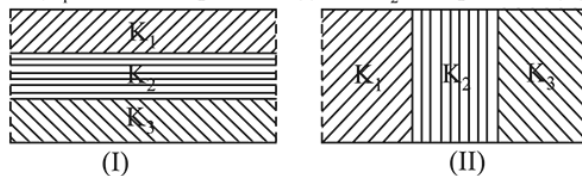
- (1) $40 \mu\text{F}$ and $10 \mu\text{F}$ (2) $60 \mu\text{F}$ and $40 \mu\text{F}$
 (3) $50 \mu\text{F}$ and $30 \mu\text{F}$ (4) $20 \mu\text{F}$ and $30 \mu\text{F}$

10 April Morning

Q14

Two identical parallel plate capacitors, of capacitance C each, have plates of area A , separated by a distance d . The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1 , K_2 and K_3 . The first capacitor is filled as shown in Fig. I, and the second one is filled as shown in Fig. II.

If these two modified capacitors are charged by the same potential V , the ratio of the energy stored in the two, would be (E_1 refers to capacitors (I) and E_2 to capacitors (II) :

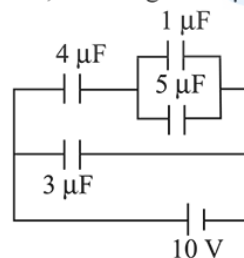


- (1) $\frac{E_1}{E_2} = \frac{K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$
 (2) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{K_1 K_2 K_3}$
 (3) $\frac{E_1}{E_2} = \frac{9 K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$
 (4) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{9 K_1 K_2 K_3}$

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Q15

In the given circuit, the charge on $4 \mu\text{F}$ capacitor will be :



- (1) $5.4 \mu\text{C}$ (2) $9.6 \mu\text{C}$ (3) $13.4 \mu\text{C}$ (4) $24 \mu\text{C}$

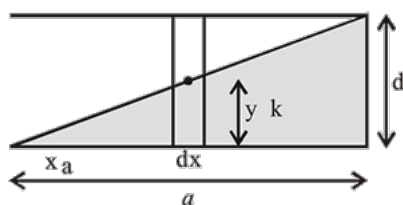
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Capacitance - Answers

Q1

(2)



$$\text{From figure, } \frac{y}{x} = \frac{d}{a} \Rightarrow y = \frac{d}{a}x$$

$$dy = \frac{d}{a}(dx) \Rightarrow \frac{1}{dc} = \frac{y}{K\epsilon_0 adx} + \frac{(d-y)}{\epsilon_0 adx}$$

$$\frac{1}{dc} = \frac{y}{\epsilon_0 abx} \left(\frac{y}{k} + d - y \right)$$

$$\int dc = \int \frac{\epsilon_0 adx}{\frac{y}{k} + d - y}$$

$$\text{or, } c = \epsilon_0 a \cdot \frac{a}{d} \int_0^d \frac{dy}{d + y \left(\frac{1}{k} - 1 \right)}$$

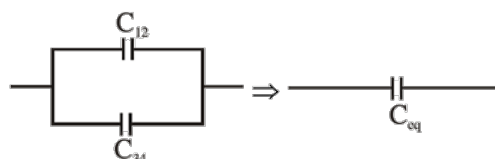
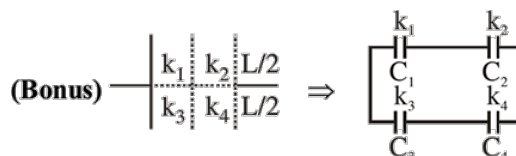
$$= \frac{\epsilon_0 a^2}{\left(\frac{1}{k} - 1 \right) d} \left[\ln \left(d + y \left(\frac{1}{k} - 1 \right) \right) \right]_0^d$$

$$= \frac{k \epsilon_0 a^2}{(1-k)d} \ln \left(\frac{d + d \left(\frac{1}{k} - 1 \right)}{d} \right)$$

$$= \frac{k \epsilon_0 a^2}{(1-k)d} \ln \left(\frac{1}{k} \right) = \frac{k \epsilon_0 a^2 \ln k}{(k-1)d}$$

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Q2



$$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{k_1 \epsilon_0 \frac{L}{2} \times L}{d/2} \cdot \frac{k_2 \left[\epsilon_0 \frac{L}{2} \times L \right]}{d/2}}{(k_1 + k_2) \left[\frac{\epsilon_0 \frac{L}{2} \times L}{d/2} \right]}$$

$$C_{12} = \frac{k_1 k_2 \epsilon_0 L^2}{(k_1 + k_2) d}$$

in the same way we get,

$$C_{34} = \frac{k_3 k_4 \epsilon_0 L^2}{(k_3 + k_4) d}$$

$$\therefore C_{eq} = C_{12} + C_{34} = \left[\frac{k_1 k_2}{k_1 + k_2} + \frac{k_3 k_4}{k_3 + k_4} \right] \frac{\epsilon_0 L^2}{d} \quad \dots (i)$$

$$\text{Now if } k_{eq} = K, C_{eq} = \frac{k \epsilon_0 L^2}{d} \quad \dots (ii)$$

on comparing equation (i) to equation (ii), we get

$$k_{eq} = \frac{k_1 k_2 (k_3 + k_4) + k_3 k_4 (k_1 + k_2)}{(k_1 + k_2)(k_3 + k_4)}$$

This does not match with any of the options so this must be a bonus.

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Q3

Capacitance

(3) Let dielectric constant of material used be K .

$$\frac{k_1 \epsilon_0 A_1}{d} + \frac{k_2 \epsilon_0 A_2}{d} + \frac{k_3 \epsilon_0 A_3}{d} = \frac{k \epsilon_0 A}{d}$$

or

$$\frac{10 \epsilon_0 A/3}{d} + \frac{12 \epsilon_0 A/3}{d} + \frac{14 \epsilon_0 A/3}{d} = \frac{K \epsilon_0 A}{d}$$

$$\frac{\epsilon_0 A}{d} \left(\frac{10}{3} + \frac{12}{3} + \frac{14}{3} \right) = \frac{K \epsilon_0 A}{d}$$

$$\therefore K = 12$$

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Q4

(2) $W = -\Delta u$

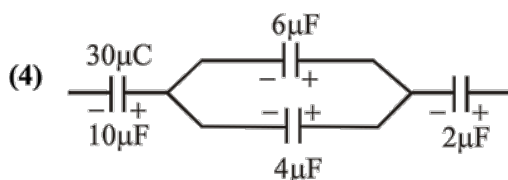
$$= (-1) \left| \frac{(c\epsilon)^2}{2kc} - \frac{(c\epsilon)^2}{2c} \right|$$

$$= \frac{\epsilon^2 c}{2} \frac{k-1}{k}$$

$$= 508 \text{ J}$$

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Q5



As given in the figure, $6\mu\text{F}$ and $4\mu\text{F}$ are in parallel.
Now using charge conservation

$$\text{Charge on } 6\mu\text{F capacitor} = \frac{6}{6+4} \times 30$$

$$= 18\mu\text{C}$$

Since charge is asked on right plate therefore is $+18\mu\text{C}$

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Q6

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(2) As required equivalent capacitance should be

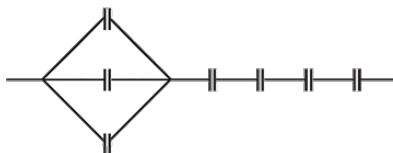
$$C_{eq} = \frac{6}{13} \mu\text{F}$$

Therefore three capacitors must be in parallel and 4 must be in series with it.

$$\frac{1}{C_{eq}} = \left[\frac{1}{3C} \right] + \left[\frac{1}{C} + \frac{1}{C} + \frac{1}{C} \right]$$

$$C_{eq} = \frac{3C}{13} = \frac{6}{13} \mu\text{F} \quad [\text{as } C = 2\mu\text{F}]$$

So, desired combination will be as below:



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Q7

(2) Energy stored in the system initially

$$U_i = \frac{1}{2} C E^2$$

$$U_f = \frac{1}{2} \frac{Q^2}{C_{eq}} = \frac{(CE)^2}{2 \times 4C} = \frac{1}{2} \frac{CE^2}{4}$$

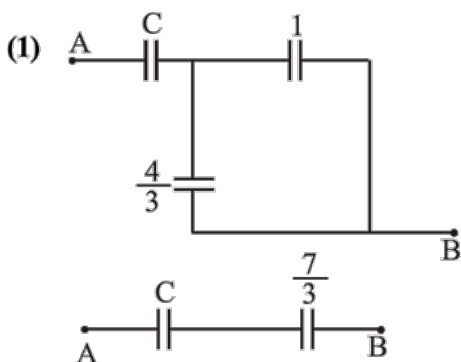
[As $Q = CE$, and $C_{eq} = 4C$]

$$\Delta U = \frac{1}{2} C E^2 \times \frac{3}{4} = \frac{3}{8} C E^2 = \frac{3}{8} \frac{Q^2}{C}$$

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Q8

Capacitance



For series combination

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow \frac{\frac{7C}{3}}{\frac{7}{3} + C} = \frac{1}{2}$$

$$\Rightarrow 14C = 7 + 3C$$

$$\Rightarrow C = \frac{7}{11} \mu\text{F}$$

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Q9

$$(3) E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$

$$\therefore Q = \epsilon_0 \cdot E \cdot A = 8.85 \times 10^{-12} \times 100 \times 1$$

$$= 8.85 \times 10^{-10} \text{C}$$

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Q10

(2) Capacitance of a capacitor with a dielectric of dielectric constant k is given by

$$C = \frac{k \epsilon_0 A}{d}$$

$$\therefore E = \frac{V}{d} \quad \therefore C = \frac{k \epsilon_0 A E}{V}$$

$$15 \times 10^{-12} = \frac{k \times 8.86 \times 10^{-12} \times 10^{-4} \times 10^6}{500}$$

$$k = 8.5$$

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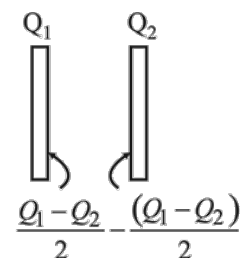
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Q11

$$(2) V = \frac{Q}{C}$$

$$= \left(\frac{Q_1 - Q_2}{2C} \right)$$

$$= \left(\frac{4 - 2}{2 \times 1} \right) = 1 \text{ V}$$



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Q11

$$(4) U = \frac{1}{4\pi\epsilon_0} \left[\frac{q(-q)}{d} + \frac{qQ}{\left(D + \frac{d}{2}\right)} + \frac{(-q)Q}{\left(D - \frac{d}{2}\right)} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{d} + \frac{qQ\left(D - \frac{d}{2}\right) - qQ\left(D + \frac{d}{2}\right)}{D^2 - \frac{d^2}{4}} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right], \because \frac{d^2}{4} \ll D$$

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Q12

$$(2) W = U_f - U_i = \frac{q^2}{2} \left(\frac{1}{C_f} - \frac{1}{C_i} \right) \left(\because U = \frac{q^2}{2C} \right)$$

$$= \frac{(5 \times 10^{-6})^2}{2} \left(\frac{1}{2} - \frac{1}{5} \right) \times 10^6$$

$$= 3.75 \times 10^{-6} \text{J}$$

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Q13

$$(4) V' = \frac{CV + (nC)V}{kC + nC}$$

$$\frac{(n+1)V}{k+n}$$

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Q14

Capacitance

- (1) Equivalent capacitance in series combination (C') is given by

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C' = \frac{C_1 C_2}{C_1 + C_2}$$

For parallel combination equivalent capacitance $C'' = C_1 + C_2$

For parallel combination

$$q = 10(C_1 + C_2)$$

$$q_1 = 500 \mu\text{C}$$

$$500 = 10(C_1 + C_2)$$

$$C_1 + C_2 = 50 \mu\text{F} \quad \dots(i)$$

For Series Combination—

$$q_2 = 10 \frac{C_1 C_2}{(C_1 + C_2)}$$

$$80 = 10 \frac{C_1 C_2}{50} \text{ From equation } \dots(ii)$$

$$C_1 C_2 = 400 \quad \dots(iii)$$

From equation (i) and (ii)

$$C_1 = 10 \mu\text{F} \quad C_2 = 40 \mu\text{F}$$

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Q14

$$(3) \frac{1}{C_1} = \frac{d/3}{k_1 \epsilon_0 A} + \frac{d/3}{k_2 \epsilon_0 A} + \frac{d/3}{k_3 \epsilon_0 A}$$

$$\text{or } C_1 = \frac{3k_1 k_2 k_3 \epsilon_0 A}{d(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

$$C_2 = \frac{k_1 \epsilon_0 (A/3)}{d} + \frac{k_2 \epsilon_0 (A/3)}{d} + \frac{k_3 \epsilon_0 (A/3)}{d}$$

$$= \frac{(k_1 + k_2 + k_3) \epsilon_0 A}{3d}$$

$$\frac{U_1}{U_2} = \frac{\frac{1}{2} C_1 V^2}{\frac{1}{2} C_2 V^2}$$

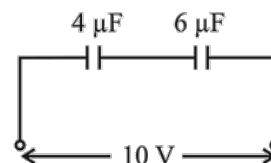
$$= \frac{E_1}{E_2} = \frac{9k_1 k_2 k_3}{(k_1 + k_2 + k_3)(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

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Q15

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- (4) $V_1 + V_2 = 10$



$$\text{and } 4V_1 = 6V_2$$

On solving above equations, we get

$$V_1 = 6 \text{ V}$$

Charge on $4 \mu\text{F}$,

$$q = CV_1 = 4 \times 6 = 24 \mu\text{C}.$$

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